

CATANIA SIGONELLA, Italy

WMO: 164590

Lat: 37.406N

Lon: 14.919E

Elev: 22

StdP: 101.06

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	1.9	3.1	-4.0	2.7	12.7	-1.9	3.2	11.3	13.9	13.2	12.5	12.8	3.1	270	0.490

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	13.0	36.9	22.2	35.1	22.2	33.8	22.1	26.5	29.4	25.6	28.8	24.7	28.2	6.0	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.0	21.4	27.6	24.9	20.0	26.7	23.9	18.8	26.2	83.0	29.9	78.6	29.3	75.1	29.1	32.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
12.0	10.4	9.3	DB	-1.0	41.4	1.3	2.2	-2.0	43.0	-2.7	44.3	-3.5	45.5	-4.5	47.1	(4)
			WB	-1.2	28.3	1.2	1.2	-2.1	29.1	-2.8	29.8	-3.5	30.4	-4.4	31.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	18.0	10.3	10.5	12.4	15.1	19.2	23.8	26.7	27.2	24.0	20.0	15.3	11.6	(6)
(7)		DBStd	6.51	2.26	2.32	2.40	2.21	2.45	2.72	2.14	1.94	2.51	2.59	2.69	2.40	(7)
(8)		HDD10.0	62	23	20	6	0	0	0	0	0	0	1	12		(8)
(9)		HDD18.3	1091	250	220	183	101	18	1	0	0	1	13	96	209	(9)
(10)		CDD10.0	2999	32	33	81	152	284	413	519	533	421	311	159	61	(10)
(11)		CDD18.3	986	0	0	1	2	44	164	261	274	171	65	4	0	(11)
(12)		CDH23.3	10180	0	0	12	37	431	1739	3104	3109	1374	354	19	1	(12)
(13)	CDH26.7	4663	0	0	3	3	106	759	1623	1590	509	68	1	0	(13)	
(14)	Wind	WSAvg	4.2	4.1	4.4	4.4	4.6	4.5	4.3	4.0	3.9	3.7	3.8	3.9	(14)	
(15)	Precipitation	PrecAvg	518	68	49	41	40	23	9	5	16	29	94	62	79	(15)
(16)		PrecMax	914	227	168	134	172	134	39	34	99	83	430	230	333	(16)
(17)		PrecMin	167	5	4	4	5	0	0	0	0	2	0	0	3	(17)
(18)		PrecStd	181	55	42	26	37	25	11	10	23	21	105	57	68	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	19.6	20.7	24.8	26.5	32.5	38.2	40.0	39.2	36.1	31.0	25.2	21.0	(19)
(20)			MCWB	13.4	13.5	14.8	16.1	18.8	21.6	21.9	22.5	21.7	20.1	18.1	15.1	(20)
(21)		2%	DB	17.9	18.2	21.8	24.1	29.7	34.8	36.9	36.8	33.1	28.2	23.7	19.1	(21)
(22)			MCWB	12.2	12.0	13.7	15.1	18.1	21.0	22.0	22.6	22.8	20.7	17.5	14.0	(22)
(23)		5%	DB	16.4	17.0	19.9	22.8	27.2	32.8	35.0	35.0	31.2	26.9	22.0	17.8	(23)
(24)			MCWB	11.7	11.7	12.8	15.0	17.3	20.8	22.4	22.6	22.3	20.1	17.1	13.1	(24)
(25)		10%	DB	15.1	15.8	18.0	20.9	25.8	30.9	33.6	33.4	29.8	25.1	20.2	16.2	(25)
(26)			MCWB	11.3	11.3	12.3	14.4	16.9	20.1	22.2	22.6	21.6	19.9	16.3	12.8	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	15.6	15.8	16.6	19.1	21.9	25.7	27.2	28.5	27.4	25.5	21.5	17.3	(27)
(28)			MCDWB	17.1	16.9	20.4	22.5	26.4	29.2	31.2	31.0	29.7	26.7	22.9	18.6	(28)
(29)		2%	WB	14.1	14.3	15.4	17.6	20.6	24.0	25.9	26.7	25.8	23.8	19.7	15.8	(29)
(30)			MCDWB	16.1	16.2	18.6	20.8	24.7	28.2	29.5	29.3	28.2	25.3	21.4	17.5	(30)
(31)		5%	WB	13.1	13.2	14.5	16.6	19.6	22.9	24.9	25.8	24.7	22.3	18.4	14.7	(31)
(32)			MCDWB	15.0	15.3	17.2	19.6	23.6	28.0	28.9	29.1	27.3	24.3	20.4	16.3	(32)
(33)		10%	WB	12.2	12.2	13.6	15.6	18.6	22.0	24.0	24.9	23.7	21.2	17.3	13.7	(33)
(34)			MCDWB	14.4	14.6	16.2	18.6	22.7	27.3	28.5	28.5	26.6	23.3	19.3	15.4	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	9.9	10.4	11.1	11.5	12.5	13.2	13.6	13.0	11.2	10.4	10.1	9.6	(35)
(36)			MCDBR	11.8	12.2	14.2	14.6	15.1	15.8	16.0	15.2	13.3	12.2	11.8	11.6	(36)
(37)			MCWBR	7.4	7.1	7.4	7.2	6.1	6.4	6.5	6.4	6.0	6.3	6.9	7.1	(37)
(38)		5% WB	MCDWB	10.1	10.2	11.5	11.6	12.3	13.6	13.4	12.5	11.7	10.1	9.9	9.4	(38)
(39)			MCWBR	7.3	6.9	7.3	7.1	6.5	6.7	6.9	6.7	6.3	6.7	7.1	7.0	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.356	0.382	0.433	0.474	0.476	0.459	0.458	0.467	0.464	0.435	0.394	0.362	(40)	
(41)		taud	2.383	2.287	2.134	2.045	2.086	2.176	2.194	2.168	2.179	2.227	2.304	2.376	(41)	
(42)		Ebn at Noon	817	838	824	811	816	828	825	809	786	769	761	780	(42)	
(43)		Edh at Noon	91	112	143	166	162	148	144	145	136	118	98	86	(43)	
(44)	All-Sky Solar Radiation	RadAvg	2.29	3.07	4.33	5.50	6.58	7.37	7.60	6.67	4.97	3.61	2.52	2.04	(44)	
(45)		RadStd	0.19	0.29	0.33	0.39	0.36	0.35	0.24	0.37	0.26	0.27	0.19	0.18	(45)	

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (8 neighbors)	N/A	N/A	N/A	N/A	-0.38	-0.49	N/A	N/A	N/A	N/A

Nomenclature: See separate page